



**Generation Interconnection
System Impact Study Report
for**

Queue Project AF2-006

JERMYN TAP-BROWNELL 69 KV

12 MW Capacity / 19.8 MW Energy

February 2021

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is PPL.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

3 General

The Interconnection Customer (IC), **Renewable Energy Aggregators**, has proposed a Solar generating facility located in Lackawanna County, Pennsylvania. The installed facilities will have a total capability of 19.8 MW with 12 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is September 16, 2021. This study does not imply a TO commitment to this in-service date.

Queue Number	AF2-006
Project Name	JERMYN TAP-BROWNELL 69 KV
State	Pennsylvania
County	Lackawanna
Transmission Owner	PPL
MFO	19.8
MWE	19.8
MWC	12
Fuel	Solar
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

AF2-006 will interconnect via the Lackawanna – East Carbondale #2 69 kV line, 2.1 miles from the Jermyn Tap 2 bus and 2.9 miles from the Brownell bus. The Point of Interconnection (POI) will be at the PPL EU owned termination structure where the Interconnection Customer’s transmission line terminates (with insulators).

5 Cost Summary

The AF2-006 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$1,228,400
Allocation towards System Network Upgrade Costs*	\$0
Total Costs	\$1,228,400

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

6 Transmission Owner Scope of Work

The total physical interconnection costs is given in the table below:

Description	Total Cost
Attachment Facilities	\$ 893,800
Direct Connection Network Upgrade	\$ 0
Non-Direct Connection Network Upgrades	\$ 334,600
Total Physical Interconnection Costs	\$ 1,228,400

PPL EU can accommodate this interconnection by constructing a new transmission line tap off the Lackawanna – East Carbondale #2 69 kV line and completing associated remote end relay work.

Risks and Assumptions

- No major environmental, real estate, siting, or permitting issues.
- IC is responsible for acquisition of easements, permits, and right of way for any Direct Connection Network Upgrades and Attachment Facilities per PPL EU standards and requirements.
- PPL EU will perform all grading, site preparation, and establish access roads for the PPL EU owned Attachment Facilities per PPL EU standards and requirements.

6.1 Attachment Facilities

69 kV Transmission Line Tap

PPL EU will tap the Lackawanna – East Carbondale #2 69 kV line at or near GPS Coordinates: 41.5305361, - 75.5149972. PPL EU will extend the tap southeast towards the IC site. PPL EU will install a motor operated switch and POI termination structure. The IC must build the remainder of the Attachment Facilities from the POI

termination structure to the IC substation. The IC is responsible for procuring 100 ft. ROW for these facilities. For the purposes of this Feasibility Study Report cost estimate, PPL EU is assuming all engineering and construction responsibility for land development activities, including grading, site preparation, and new access road. During the Facilities Study phase, PPL EU and the IC will review land development activities, and the IC may choose to perform some, or all, of these activities. The cost estimate will be updated accordingly and included in the Facilities Study Report.

PPL EU work will consist of the following:

- Install one (1) new double circuit, tension, monopole, custom steel/foundation, tap structure.
- Install one (1) new single circuit, direct-embed, steel, motor operated switch structure.
- Install one (1) new single circuit, direct-embed, steel, tension structure.
- New circuit will consist of three (3) phase conductors and one (2) shield wire (OHGW). The total preliminary cost estimate for the Attachment work is given in the table below.

Description	Total Cost
69 kV Transmission Line Tap	\$ 869,400
IC substation facility acceptance, testing, commissioning, & telemetry coordination	\$ 24,400
Total Attachment Facility Costs	\$ 893,800

6.2 Direct Connection Cost Estimate

None

6.3 Non-Direct Connection Cost Estimate

Remote End Relay Work – Lackawanna 69 kV Substation

- Complete remote end relay work at Lackawanna 69 kV substation for Direct Transfer Trip.
- Model IC in CAPE and conduct a wide area short-circuit study two busses away from the IC facilities. Identify affected relays and revise settings as needed.
- Conduct a review of the IC relay settings and engineering package (submitted by IC to PPL EU).
- PPL EU does not have optical ground wire (OPGW) available on the Lackawanna – East Carbondale #2 69 kV line as the communication circuit for DTT to the Lackawanna 69 kV substation. PPL EU assumes that the IC will procure the communication circuit through a third-party provider. Upon request, PPL EU will evaluate installing OPGW on the Lackawanna – East Carbondale #2 69 kV line for DTT.

Lackawanna – East Carbondale #2 69 kV Modifications to tie in the AF2-006 Attachment Facilities

- Tie the new AF2-006 Attachment Facilities into the Lackawanna – East Carbondale #2 69 kV line.

- Reframe (2) existing H-frame to steel, direct-embed, tension structures on the Lackawanna – East Carbondale #2 69 kV line on either side of the tap.

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below.

Description	Total Cost
Remote End Relay Work – Lackawanna 69 kV Substation	\$ 238,000
Lackawanna – East Carbondale #2 69 kV Modifications to tie in the AF2-006 Attachment Facilities	\$ 96,600
Total Non-Direct Connection Facility Costs	\$ 334,600

7 Schedule

The estimated time to complete the scope of work is 12-18 months after the PJM three-party Interconnection Service Agreement (ISA) and Interconnection Construction Service Agreement (ICSA) are signed and PPL EU receives Notice to Proceed from the IC.

8 Interconnection Customer Requirements

PPL EU applicable technical standards that address requirements for interconnection of generation, transmission, and end user facilities can be found at the following link:

<https://pjm.com/planning/design-engineering/to-tech-standards/private-ppl.aspx>

9 Revenue Metering and SCADA Requirements

9.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

9.2 Meteorological Data Reporting Requirements

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Temperature (degrees Fahrenheit)
- Atmospheric pressure (hectopascals)
- Irradiance
- Forced outage data

9.3 Interconnected Transmission Owner Requirements

The IC will be required to comply with all Interconnected Transmission Owner's revenue metering requirements for generation interconnection customers located at the following link:

<http://www.pjm.com/planning/design-engineering/to-tech-standards/>

10 Summer Peak Analysis

The Queue Project AF2-006 was evaluated as a 20 MW (Capacity 12 MW) injection tapping the Jermyn to Brownell 69 kV line in the PPL area. Project AF2-006 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF2-006 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

10.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

10.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

10.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

10.4 Steady-State Voltage Requirements

To be determined during the Facilities Study as applicable

10.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection

Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
98990902	208009	LACK	230	PPL	200074	LACKAW	500.0	PJM	3	PL_P13_100348	operation	1167	99.98	100.67	AC	8.35
98990903	208009	LACK	230	PPL	200074	LACKAW	500.0	PJM	3	PL_P13_100328	operation	1167	99.98	100.67	AC	8.35

10.6 Contingency Descriptions

Contingency Name	Contingency Definition
PL_P13_100348	CONTINGENCY 'PL_P13_100348' /* LACK 500/230KV T4 DISCONNECT BRANCH FROM BUS 200074 TO BUS 208009 CKT 4 /* LACK-LACKAW 500-230 T3 END
PL_P13_100328	CONTINGENCY 'PL_P13_100328' /* LACK 230/69KV T2 DISCONNECT BRANCH FROM BUS 211681 TO BUS 208009 CKT 2 /* LACK T2 DISCONNECT BRANCH FROM BUS 200074 TO BUS 208009 CKT 4 /* LACK T4 END

11 Light Load Analysis

The Queue Project AF2-006 was evaluated as a 20.0 MW injection tapping the Jermyn to Brownell 69 kV line in the PPL area. Project AF2-006 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF2-006 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

11.1 Light Load Deliverability

(Single or N-1 contingencies)

None

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies)

None

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

11.4 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

11.5 System Reinforcements

None

12 Short Circuit Analysis

The following Breakers are overdutied:

None

12.1 System Reinforcements - Short Circuit

None

13 Stability and Reactive Power

(Summary of the VAR requirements based upon the results of the dynamic studies)

To be determined in the Facilities Study Phase.

14 Affected Systems

14.1 NYISO

NYISO Impacts to be determined during later study phases (as applicable).

15 Attachment 1: One Line Diagram

